

P11-B DE RUYTER, NetherlandsWMO: **062030**Lat: **52.360N**Lon: **3.342E**Elev: **24**StdP: **101.04**Time Zone: **1.00 (EUC)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.2	0.1	-7.5	2.0	0.1	-5.4	2.4	1.2	21.1	8.1	19.2	7.9	9.4	60	0.789

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.1	21.4	18.2	20.1	17.4	19.2	17.0	18.8	20.6	18.1	19.6	17.5	18.9	6.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	20.0	17.4	12.4	19.2	16.8	12.0	18.6	53.5	20.6	51.2	19.7	49.3	19.0	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	16.1	14.5	-1.2	25.4	2.3	2.5	-2.9	27.2	-4.2	28.7	-5.6	30.1	-7.2	32.0		
(5)			-3.2	19.9	2.6	1.3	-5.0	20.9	-6.6	21.6	-8.0	22.4	-9.9	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	6.1	5.3	6.6	8.8	11.2	14.0	16.6	17.6	16.2	13.8	10.2	7.9
	DBStd	4.72	2.58	2.75	2.27	2.33	2.29	1.99	1.78	1.46	1.69	2.04	2.36	2.62
	HDD10.0	522	122	132	106	50	13	1	0	0	0	1	25	72
	HDD18.3	2617	381	365	363	287	221	132	61	32	66	141	246	323
	CDD10.0	964	0	0	1	13	50	120	204	234	187	119	29	7
	CDD18.3	19	0	0	0	0	0	2	6	8	3	0	0	0
	CDH23.3	18	0	0	0	0	0	2	12	2	2	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	7.5	8.9	8.7	7.5	6.7	6.3	6.1	5.9	6.5	6.9	8.0	8.5	9.6
Precipitation	PrecAvg	908	85	71	62	51	64	73	79	86	71	78	90	99
	PrecMax	1142	183	160	141	115	133	131	149	200	198	156	146	216
	PrecMin	716	6	19	7	1	19	19	22	18	16	22	14	25
	PrecStd	111	38	32	31	27	26	31	34	46	39	29	33	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	11.1	12.6	17.0	18.5	21.6	25.0	23.1	22.2	19.1	15.9	13.1
		MCWB	9.8	8.8	9.5	11.6	13.8	17.4	20.1	18.5	19.1	16.5	14.3	11.6
	2%	DB	10.4	9.7	10.6	14.3	16.5	19.0	21.1	21.2	20.1	18.0	15.0	12.5
		MCWB	9.2	7.9	8.6	10.8	13.4	16.5	18.0	18.4	17.9	16.1	13.4	11.1
	5%	DB	9.9	9.0	9.8	12.5	15.0	17.1	19.6	20.0	19.1	17.2	14.4	11.9
		MCWB	8.7	7.4	8.2	10.2	12.8	15.3	17.3	17.4	16.9	15.4	12.7	10.6
	10%	DB	9.2	8.4	9.1	11.4	14.0	16.1	18.6	19.2	18.3	16.5	13.5	11.1
		MCWB	7.9	7.0	7.6	9.7	12.4	14.5	16.7	17.1	16.3	14.6	11.8	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	9.4	9.7	12.0	15.2	18.1	19.7	19.6	19.6	17.0	14.7	12.0
		MCDB	10.7	10.3	11.7	15.8	16.8	20.8	22.8	21.6	21.6	18.3	15.6	12.7
	2%	WB	9.6	8.2	9.1	11.1	13.8	16.7	18.6	18.7	18.6	16.3	13.7	11.4
		MCDB	10.2	9.2	10.4	13.3	15.1	18.6	20.5	20.3	19.9	17.8	14.8	12.2
	5%	WB	8.9	7.7	8.5	10.5	13.1	15.5	17.8	18.1	17.5	15.7	13.0	10.8
		MCDB	9.7	8.7	9.7	12.3	14.5	16.9	19.3	19.5	18.7	17.1	14.4	11.8
	10%	WB	8.2	7.2	7.9	9.9	12.4	14.7	17.0	17.5	16.8	14.9	12.1	9.8
		MCDB	9.1	8.3	8.9	11.3	13.9	15.8	18.2	19.0	18.1	16.3	13.5	10.9
Mean Daily Temperature Range	5% DB	MDBR	2.4	2.3	2.2	2.5	2.2	2.1	2.1	2.1	2.2	2.4	2.4	2.4
		MCDBR	2.8	2.4	2.9	4.5	4.2	4.1	3.8	3.1	2.5	2.7	2.9	2.7
		MCWBR	3.3	2.2	2.3	2.4	2.5	2.9	2.4	2.3	2.5	2.7	3.3	3.3
	5% WB	MCDBR	2.8	2.4	2.6	3.9	3.5	3.9	3.4	2.6	2.3	2.8	2.8	2.6
		MCWBR	3.4	2.3	2.3	2.3	2.5	3.0	2.5	2.5	2.6	2.9	3.4	3.4
Clear-Sky Solar Irradiance	taub	0.334	0.351	0.383	0.406	0.411	0.409	0.414	0.405	0.388	0.367	0.343	0.326	
	taud	2.354	2.372	2.305	2.279	2.332	2.352	2.348	2.386	2.422	2.425	2.381	2.342	
	Ebn at Noon	651	756	801	828	842	846	834	821	791	727	634	588	
	Edh at Noon	62	80	103	119	119	117	116	106	92	76	60	53	
All-Sky Solar Radiation	RadAvg	0.69	1.41	2.71	4.51	5.45	5.80	5.53	4.67	3.32	1.81	0.83	0.52	
	RadStd	0.06	0.25	0.28	0.40	0.34	0.31	0.58	0.33	0.22	0.17	0.08	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page